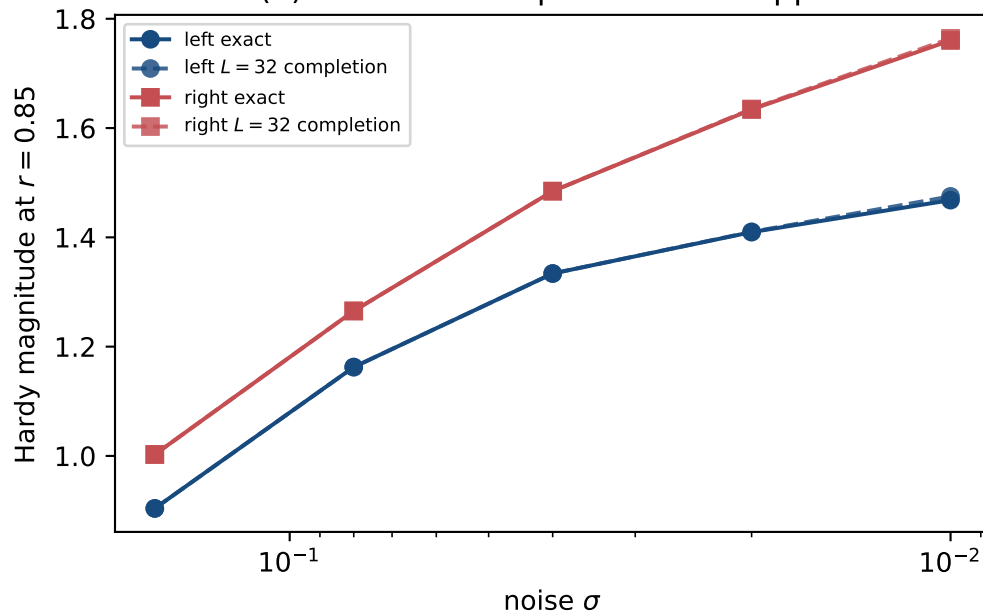
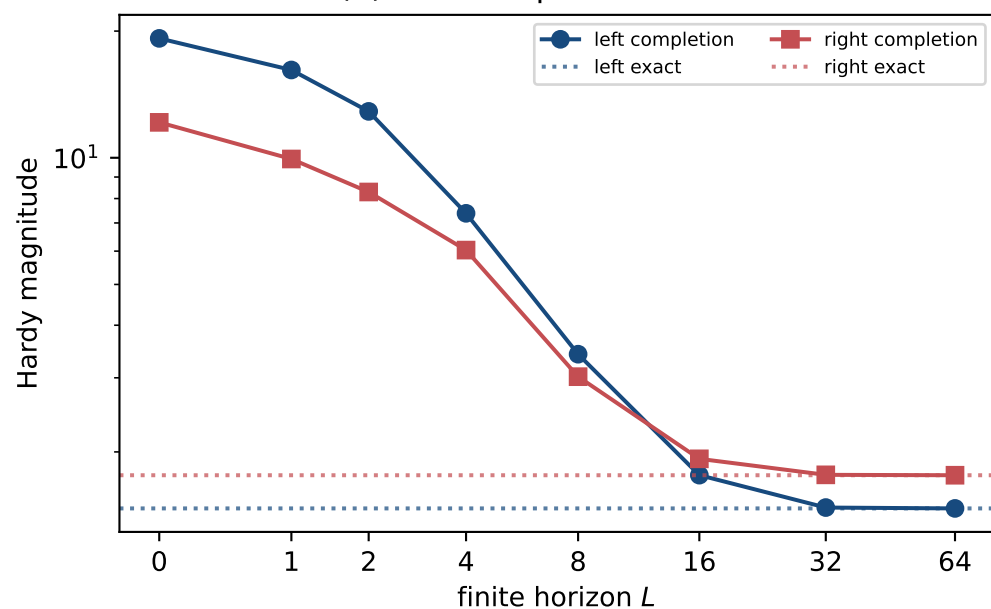
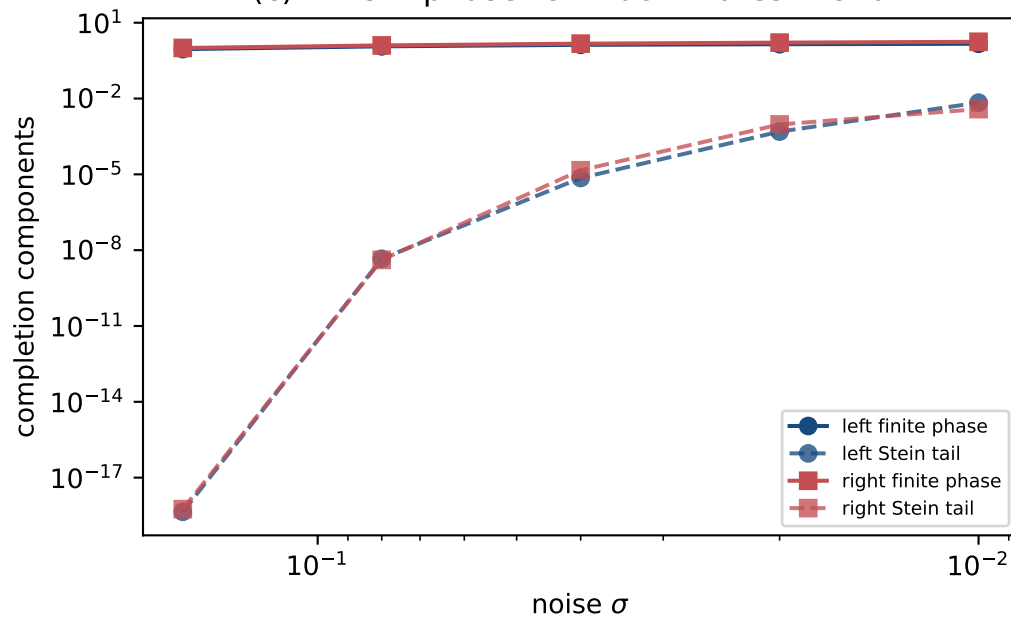


(a) Fixed-horizon phase-aware upper

(b) Tail collapse at $\sigma = 0.01$ (c) $L = 32$: phase term dominates the tail

(d) Completion removes the endpoint wall

